thing; managing such downloads for millions of users is another. When data speeds are beyond what a single desktop PC would normally be able to process, copper wires don't serve well. This is where fibreoptics come in. Fibre optics work on the theory of total internal reflection and are much faster than the traditional medium of data transmission. This is because of the medium itself, viz light.

Optical data transmission promises the highest transfer rates with least noise and power loss. Both these properties make fibre optics THE data carrier medium for inter and intra-continental networks. Apart from being used in backbone networks, fibre optics are used within data centres to connect computers which need to serve as much data as possible. Since disk speeds are usually much faster than traditional electric wire based networks, they don't do justice to the hardware capacities of these machines. Hence, fibre optics!

Flash Memory

The amount of data we own is growing at an alarming rate. It needs to be stored somewhere secure and portable. Flash memory devices do just that. Flash Memory is a form of non-volatile memory. It can be arranged and reprogrammed at will. Data in it is arranged in units called blocks.

The concept of flash memory was developed at Toshiba by Dr. Fujio Masuoka in 1980. The term 'flash' was given as this sort of storage is easily and electronically erasable. There are two types – NAND Flash memory and NOR Flash memory, the difference lying in interface for reading and writing to the device.

Flash memory is used in cameras, cell phones, LAN switches, set-top boxes, embedded controllers, MP3 players and a variety of other devices.

It was not so long ago we were using CDs and floppy disks for data storage and transfer. Inerasable and bulky, they were the only option at the